

Professional Software Development – 16 Week Course

Instructor Information

Instructor: Anthony Ackah-Nyanzu (Tony)

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Office Hours: Thursdays & Fridays (by pre-arrangement)

Course Credits: 4.0 (Prereq: CSCI 240 or equivalent)

Class Meeting Time: Mondays, 7:30 – 10:20 PM

Course Description

Practical industrial software development, including mastering and contributing to existing large codebases, applying software design principles and design alternatives, using industry-standard tools for source control, release management, testing, documentation, and DevOps, and collaborating effectively in a team environment on real-world inspired projects.

Course Goals

- Understand and critique the software design of existing systems, including trade-offs and alternatives, with instructor assistance as needed.
- Navigate, modify, and create software projects, including extending a large-scale codebase and building their own project from scratch.
- Use standard development tools (Git, npm, CI/CD, issue tracking, Postman) effectively in real-world development.
- Learn new technologies through guided self-directed study, supported by instructor guidance.
- Apply Agile and DevOps practices to plan, implement, release, and maintain software projects in a collaborative team environment.

Required & Supplied Materials

Required:

- Personal computer (Mac, Windows, or Unix). Students should bring their laptops to class if they have one.

Supplied by Department:

- Access to departmental computing resources (servers, cloud accounts, etc.)

Tools & Technologies Used in Course:

- Git & GitHub
- TypeScript and Node.js
- React
- SQL Server (primary) with PostgreSQL and MongoDB discussed as alternatives
- npm (package manager)
- Postman (for API testing)
- CI/CD with GitHub Actions
- Docker (introductory)
- Microsoft Azure (cloud deployment and DevOps)

Week-by-Week Curriculum

Week 1: Introduction to Software Engineering & Tools Setup – SDLC overview, environment setup (Git, IDE, SQL Server, Azure). Assignment: Select project idea.

Week 2: Git & GitHub Workflows + Project Confirmation + Agile Planning – Git basics, project idea confirmation, Agile planning with Azure Boards/GitHub Projects.

Week 3: APIs – Building & Consuming – REST API fundamentals, designing endpoints, consuming third-party APIs, Postman testing.

Week 4: Databases & Persistence – SQL Server CRUD, ORMs, PostgreSQL/MongoDB alternatives, migrations. Project Task: persistence layer.

Week 5: Software Design Principles – SOLID, design patterns, modularity. Project Milestone: Refactor early project code.

Week 6: Testing & Bug Tracking – Unit/integration testing, mocking, test coverage, bug tracking tools. Project Milestone: Add automated tests.

Week 7: CI/CD & DevOps Basics (with Azure) + Project Review – GitHub Actions pipelines, deployment to Azure, linting, monitoring. Project Review #1.

Week 8: Frontend Fundamentals (TypeScript + React) – React components, state management, connecting frontend to backend, deploy on Azure. Project Task: start presentation layer.

Week 9: Authentication & Authorization with Azure – AuthN vs AuthZ, Azure AD, Key Vault. Project Task: add auth.

Week 10: Working in Large Codebases + Async vs. Sync Projects – Code walkthroughs, sync vs async, message queues. Project Task: async analysis.

Week 11: Advanced Software Design & Architecture – Monolith vs microservices, REST/GraphQL, architecture refinements. Project Review #3.

Week 12: Data Warehousing, Analytics & AI Integrations – Data warehousing, SQL Server analytics, Azure Synapse, AI APIs. Project Task: analytics/AI feature.

Week 13: Release Management & DevOps – Semantic versioning, job scheduling, logging/monitoring, rollback. Project Task: scheduled job.

Week 14: Cloud Basics & Containers (Docker + Azure) – Docker, AKS, cost management. Project Task: connect frontend & API and deploy.

Week 15: Capstone Project Development & Debugging Sprint – Integration, debugging, QA. Project Review #4.

Week 16: Final Presentations, Reflection, & Exam – Project demos, peer reviews, reflection. Final Exam = Project Presentation.

Assessment & Grading

- Weekly Work (40%) – At least one GitHub pull request each week + coding exercises.
- Oral Exams & Code Reviews (30%) – Periodic oral exams and walkthroughs of code/design decisions.
- Final Exam – Capstone Project Presentation (30%) – Project demo in Week 16.

Academic Policies

- Attendance expected for all synchronous sessions.
- Weekly pull requests and milestone reviews required for credit.
- Late work requires prior arrangement.
- Students are expected to bring laptops to class if they have one.
- Academic honesty: individual projects must be completed independently; team projects are limited to assigned members.